

Morphology of retroperitoneal vascular corridor for ATP access to L2-S1 intervertebral disc:

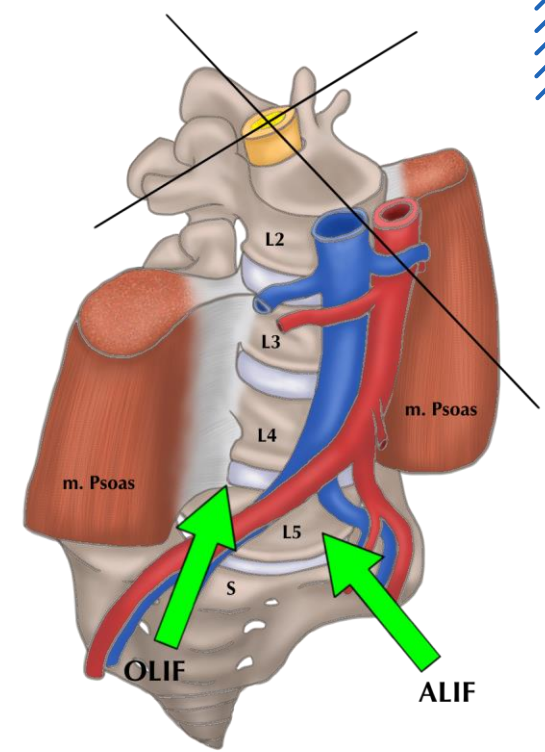
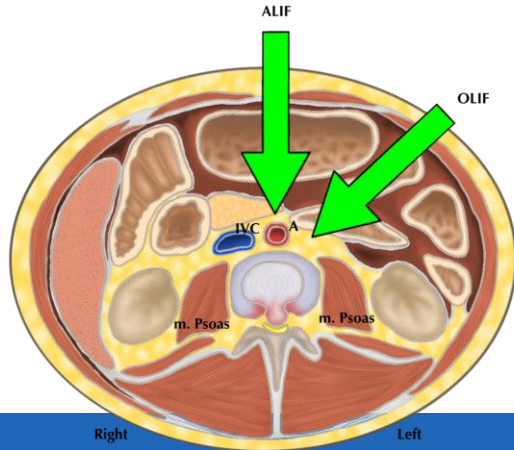
MRI study with clinical correlation

Fatmawati General Hospital
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Lumbar Interbody Fusion

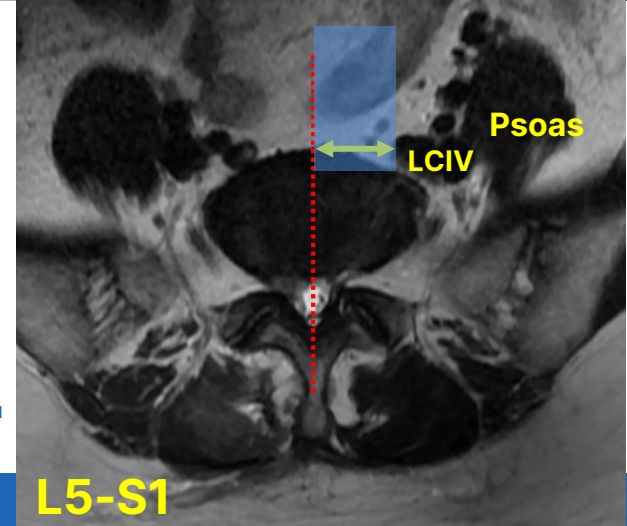
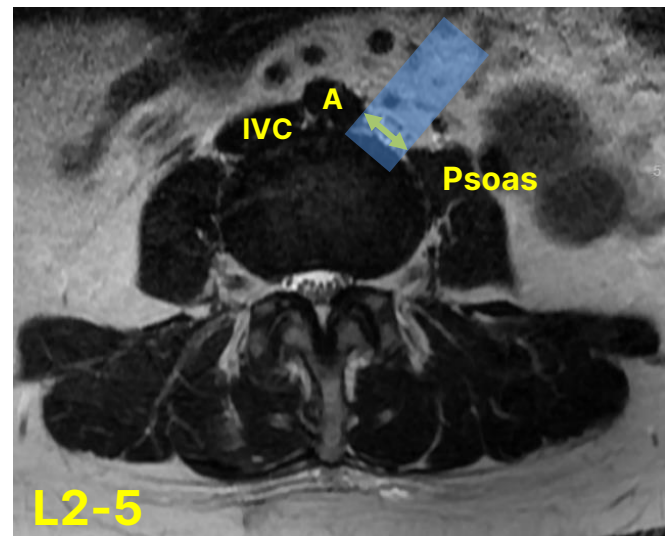
- Anterior approach LIF gained popularity
 - Adequate endplate preparation + Larger cage = Better fusion
 - Stronger correction power
 - Indirect decompression



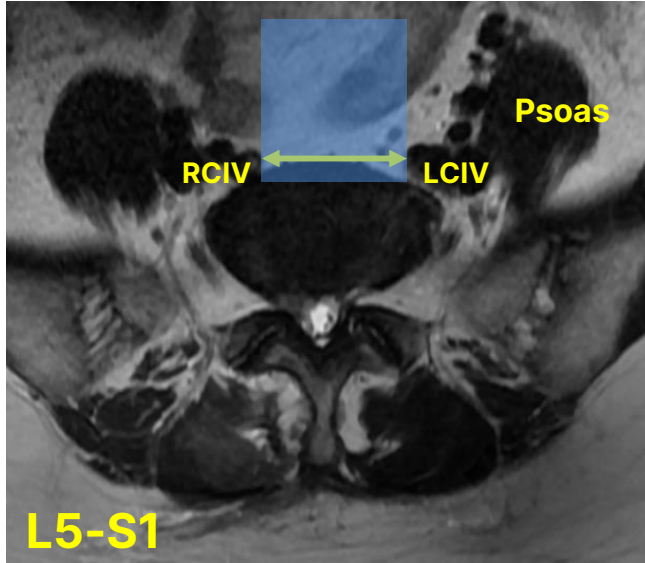
- Risk of vascular injury → OLIF 1,6-4,3%; ALIF 3% → overall 1,9%
- Approach related → fail to identify major vessel clearly

Method

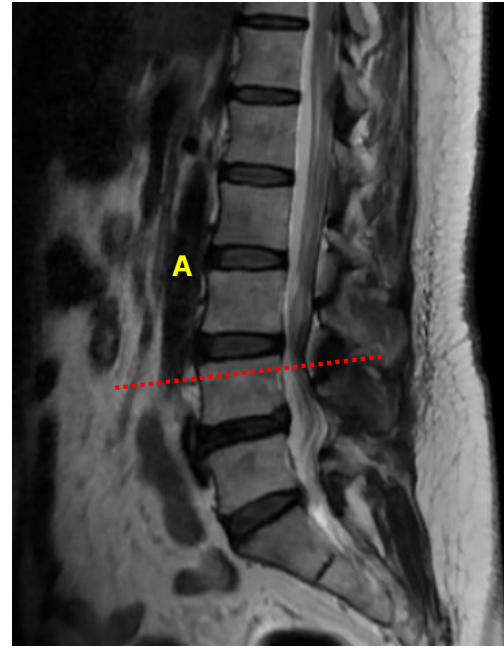
- 150 lumbar supine MRI; sagittal and axial T2WI
- Criteria:
 - LBP/sciatica patients
 - Degenerative disc/Mild spondylolisthesis
 - No deformity/scoliosis
 - No tumor/infection
 - No previous surgery/instrumentation
- ANTERIOR and OBLIQUE corridor L2-S1 disc



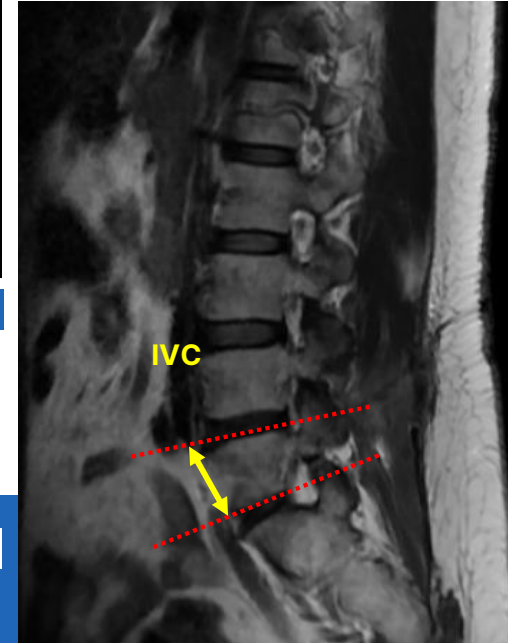
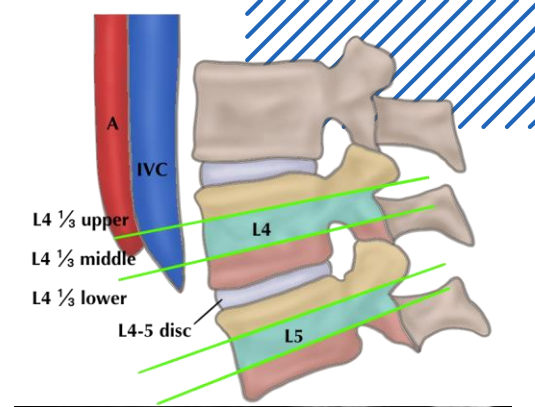
1. Oblique Corridor



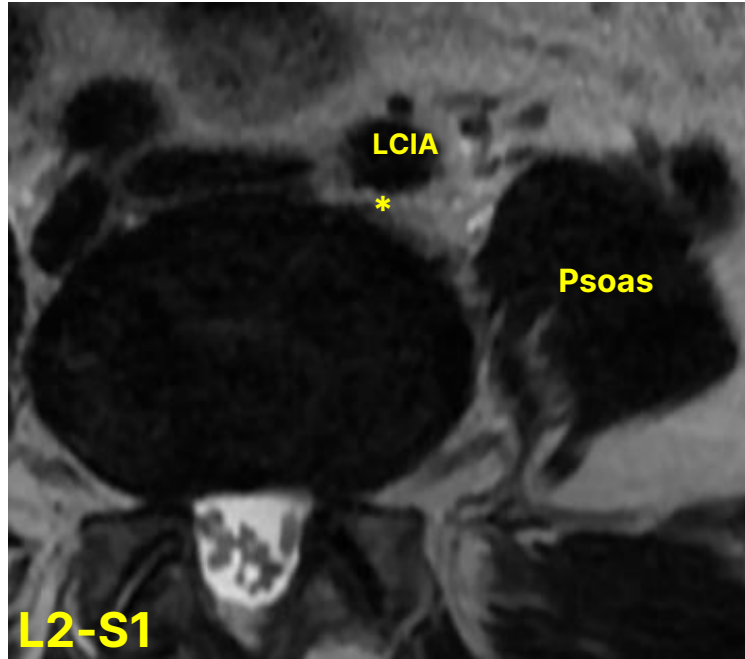
2. Anterior Horizontal Corridor



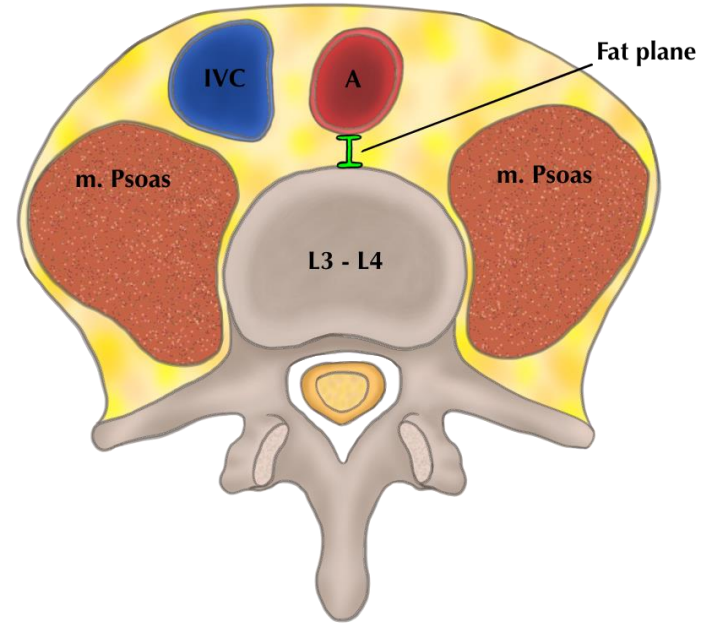
3. Bifurcation Level



4. Anterior Vertical Corridor

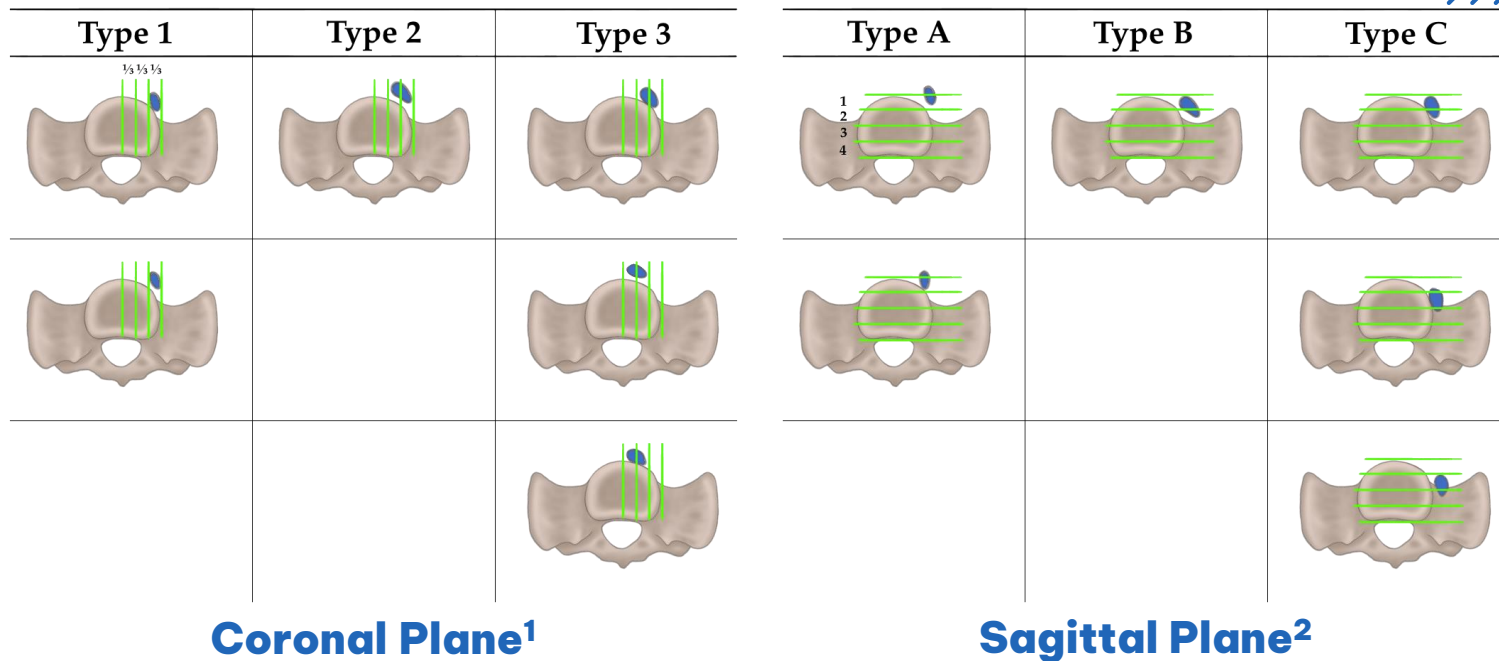


5. Perivascular Fat Plane



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6. LCIV configuration on L5-S1 disc



- To determine vessel position and difficulty for mobilization

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